

4.2.2 Haloalkanes

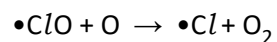
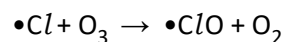
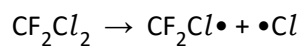
Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Substitution reactions of haloalkanes	
(a) hydrolysis of haloalkanes in a substitution reaction: (i) by aqueous alkali (ii) by water in the presence of AgNO_3 and ethanol to compare experimentally the rates of hydrolysis of different carbon–halogen bonds (b) definition and use of the term <i>nucleophile</i> (an electron pair donor) (c) the mechanism of nucleophilic substitution in the hydrolysis of primary haloalkanes with aqueous alkali (see also 4.1.1 h–i) (d) explanation of the trend in the rates of hydrolysis of primary haloalkanes in terms of the bond enthalpies of carbon–halogen bonds (C–F, C–Cl, C–Br and C–I)	PAG7 (see also 6.3.1 c) HSW1,2 Use of reaction mechanisms to explain organic reactions.

Environmental concerns from use of organohalogen compounds

- (e) production of halogen radicals by the action of ultraviolet (UV) radiation on CFCs in the upper atmosphere and the resulting catalysed breakdown of the Earth's protective ozone layer, including equations to represent:

- (i) the production of halogen radicals
- (ii) the catalysed breakdown of ozone by $\text{Cl}\cdot$ and other radicals e.g. $\cdot\text{NO}$.

Simple equations of the breakdown process are required, e.g.



Learners could be expected to construct similar equations for other stated radicals.

HSW9,10,11,12 Benefits of CFCs; acceptance of scientific evidence explaining ozone depletion leading to government legislation against CFC use.